



Facecon: Genotype-to-3D-Face Prediction System

Project Proposal

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Abstract

Reconstructing a 3D face from DNA is still an open problem. Difface showed that AI can map genetic SNPs to 3D facial meshes [1], but it was trained on private Chinese data with an unpublished SNP list. Critics also argue that its results may be driven more by age, gender, and ancestry than by true individual genetics [2]. Because the data and methods are not fully open, other teams cannot easily check or reuse the work.

This project will build a new SNP-to-3D-face system on the 3DFN dataset: dbGaP genotypes paired with FaceBase 3D face scans. We will use our own model design and a shared FLAME face template [3], instead of copying Difface's fixed 7,842-SNP setup and unknown mesh format [1]. FaceBase scans will be fitted to FLAME so every face uses the same mesh structure. The model will predict FLAME shape from genotype. We will also compare our results with simple demographic baselines, as suggested by recent critiques of Difface [2].

1. Introduction

Facial shape is partly controlled by genetics. Being able to predict a face from DNA would be useful in forensics, anthropology, and medical genetics [4]. Older methods use GWAS and simple statistical models on facial landmarks or shape scores [5]. Newer AI methods learn a shared space for DNA and face data, then generate a face from genotype.

Difface is the best-known recent system for this task [1]. It first aligns SNP and face features, then uses a diffusion model to reconstruct a 3D mesh. It was trained on about 9,674 Han Chinese subjects with 7,842 GWAS SNPs. The paper and code do not share the SNP list, preprocessing steps, face template, or trained weights. A later comment paper raises concerns about fairness of evaluation and whether the model truly captures individual genetic signal [2].

We will use 3DFN instead: controlled genotype data from dbGaP and 3D scans from FaceBase. Difface's SNP list and face template do not match this dataset, so we will build our own pipeline rather than try to reuse Difface's code [1]. For 3D faces, we will convert FaceBase scans into the FLAME format [3], which gives every subject the same mesh layout for training and output.

2. Goals and Objectives

- Choose and document an SNP panel after quality control, guided by facial GWAS findings [5].
- Fit FaceBase scans to the FLAME template [3].
- Train Stage 1: align SNP and face features in a shared space [1].
- Train Stage 2: generate FLAME face shape from SNP features [1], [3].
- Evaluate with mesh error, ranking tests, and demographic baselines [2].
- Deliver code, working system demo, method report, and results for the thesis.

3. Scope of the Project

This project covers the full path from data to evaluation. We will use 3DFN subjects who have both dbGaP genotypes and FaceBase 3D scans. Genotypes will be cleaned and reduced to a documented SNP panel [5]. Scans will be registered to FLAME [3] so each person is represented by one common mesh. We will then train a two-stage model inspired by Difface [1]. Stage 1 uses a transformer (or MLP) SNP encoder and a mesh face encoder (SpiralNet or similar), trained with CLIP-style contrastive loss so the same person's DNA and face features sit close together. Stage 2 uses a diffusion model to map SNP features to face features, then a FLAME decoder to output a 3D face (mesh vertices or FLAME shape parameters) [3]. We will measure how close the predicted face is to the real face, and compare against baselines that use only age, gender, or ancestry [2]. If sample size allows, we will also check results by ancestry, gender, and age groups. Final outputs will be preprocessing scripts, trained models, evaluation figures, a working system demo, and a written thesis.

4. Initial Study and Work Done So Far

We reviewed the Difface paper and code to decide whether to reuse it or build a new system for 3DFN [1]. Difface needs a fixed 7,842-SNP input and a dense face mesh of about 46,794 vertices. The public code is incomplete. There is no SNP list, no template mesh, no trained weights, and several hard-coded paths that break outside the authors' machine. Published critique also warns that evaluation may overstate individual-level accuracy [2]. For these reasons, we will not reuse Difface's trained weights. We will keep only the two-stage idea and design our own system for 3DFN. We have chosen FLAME as the face format [3]. We have already applied for data access, and the next steps are checking FaceBase file formats for FLAME fitting, drafting SNP selection rules [5], and testing a small Stage 1 model before full training.

References

- [1] M. Jiao et al., "De novo reconstruction of 3D human facial images from DNA sequence," *Advanced Science*, 2025. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12362825>
- [2] J. K. Wagner, N. Claessens, C. M. Maloney, and P. Claes, "Comment on 'De novo reconstruction of 3D human facial images from DNA sequence'," *Advanced Science*, 2025. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12931204>
- [3] FLAME. [Online]. Available: <https://flame.is.tue.mpg.de/>
- [4] C. Wang et al., "Forensic DNA phenotyping: a 19-SNP prediction system," *Forensic Sciences Research*, 2026. [Online]. Available: <https://doi.org/10.3788/tfsr20230045>
- [5] Z. Xiong et al., "Combined GWAS of facial traits in Europeans," *Nature Communications*, 2025. [Online]. Available: <https://www.nature.com/articles/s41467-025-61761-7>

[6] M. L. Marazita, S. Weinberg, and Z. Raffensperger, “3D Facial Norms Database,” FaceBase Consortium. [Online]. Available: https://www.facebase.org/resources/human/facial_norms/

[7] National Center for Biotechnology Information, “Genetic Analysis of Normal Human Facial Variation,” dbGaP, accession no. phs000949.v1.p1. [Online]. Available: https://www.ncbi.nlm.nih.gov/projects/gap/cgi-bin/study.cgi?study_id=phs000949.v1.p1